

Code-Layout, Readability and Reusability

Date	08 October 2026
Team ID	UNM1611BHAVANA
Project Name	AI FitTrack API
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	2-space indentation in all JavaScript files
2	Proper File Structure	Files and folders are logically organized	Yes	Separate routes/, controllers/, models/, middleware/, config/ and tests/ folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	camelCase names such as userId, workoutList, calorieTotal
4	Function / Method Names	Functions are descriptively named	Yes	Verb-based names such as registerUser, createWorkout, getProgressSummary
5	Code Comments	Inline and block comments explain logic	Yes	Block comments on controllers and middleware; inline comments on AI logic
6	Modular Design	Code is split into reusable functions/modules	Yes	Business logic in controllers, schemas in models, shared middleware
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated logic moved into shared middleware and helpers
8	Error Handling	Exceptions and errors are handled gracefully	Yes	try/catch in controllers plus centralized error-handling middleware

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication middleware (JWT verify)	JavaScript / Express	All protected routes	High
2	Centralized error handler	JavaScript / Express	Every route and controller	High
3	Validation middleware	JavaScript / express-validator	Register, login, workout, meal and goal routes	High
4	Database config (config/db.js)	JavaScript / Mongoose	server.js and test setup	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
5	AI recommendation helper	JavaScript	Workout and diet recommendation endpoints	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear layered MVC-style structure
Readability	4	Descriptive names and consistent formatting
Reusability	4	Shared middleware and helpers used across modules
Documentation / Comments	4	README plus comments on key logic
Overall Score	4.25	Well organized and maintainable